

LAB # 2

File Access Permission in LINUX

Objective

To study the various File Access Permission and different types users in LINUX.

Lab task

1. Create multiple files and perform all commands.

```
stewiedev@DESKTOP-RH97L4T:~$ mkdir lab2 && cd lab2
stewiedev@DESKTOP-RH97L4T:~/lab2$ touch file1 file2 file3
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat > file1
This will be content of the firstfilestewiedev@DESKTOP-RH97L4T:~/lab2$ cat > file2
This will be the content of the file2
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls
file1 file2 file3
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls -l
total 8
-rw-r--r-- 1 stewiedev stewiedev 37 Mar 30 00:48 file1
-rw-r--r-- 1 stewiedev stewiedev 38 Mar 30 00:49 file2
-rw-r--r-- 1 stewiedev stewiedev 0 Mar 30 00:47 file3
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls -a
. .. file1 file2 file3
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat file1
This will be content of the firstfilestewiedev@DESKTOP-RH97L4T:~/lab2$
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat file1 > file3
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat file3
This will be content of the firstfilestewiedev@DESKTOP-RH97L4T:~/lab2$
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat file1 file2 > merged.txt
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat merged.txt
This will be content of the firstfileThis will be the content of the file2
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat file1 >> file3
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat file3
This will be content of the firstfileThis will be content of the firstfilestewiedev
@DESKTOP-RH97L4T:~/lab2$ cat file1
This will be content of the firstfilestewiedev@DESKTOP-RH97L4T:~/lab2$
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls wrongfile 2> error.txt
stewiedev@DESKTOP-RH97L4T:~/lab2$ cat error.txt
ls: cannot access 'wrongfile': No such file or directory
```

2. Remove non empty directory.

```
stewiedev@DESKTOP-RH97L4T:~/lab2$ mkdir testdir
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls
error.txt file1 file2 file3 merged.txt testdir
stewiedev@DESKTOP-RH97L4T:~/lab2$ touch testdir/DummyFile
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls
error.txt file1 file2 file3 merged.txt testdir
stewiedev@DESKTOP-RH97L4T:~/lab2$ rm -r testdir
stewiedev@DESKTOP-RH97L4T:~/lab2$ ls
error.txt file1 file2 file3 merged.txt
```

3,4,5.date (for Current Date). ps(running processes),finger,Display calendar and date simultaneously.

Displaying dat and cal simultanously,
Free command

```
stewiedev@DESKTOP-RH97L4T:~/lab2$ date
Mon Mar 30 01:00:45 PKT 2026
stewiedev@DESKTOP-RH97L4T:~/lab2$ ps
  PID TTY          TIME CMD
 3481 pts/0    00:00:00 bash
 4148 pts/0    00:00:00 ps
stewiedev@DESKTOP-RH97L4T:~/lab2$ finger
Login      Name            Tty      Idle  Login Time   Office   Office Phone
stewiedev          *pts/1      1:00   Mar 30 00:01
stewiedev@DESKTOP-RH97L4T:~/lab2$ cal
  March 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31

stewiedev@DESKTOP-RH97L4T:~/lab2$ free
              total        used        free      shared  buff/cache   available
Mem:           3947764        441332        3493812          3560        157428        3506432
Swap:          1048576           0          1048576
stewiedev@DESKTOP-RH97L4T:~/lab2$ date +%Y > yearfile
cat yearfilestewiedev@DESKTOP-RH97L4T:~/lab2$ cat yearfile
2026
stewiedev@DESKTOP-RH97L4T:~/lab2$ date && cal
Mon Mar 30 01:05:45 PKT 2026
  March 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31
```

3. df./
4. df-k./
5. du -s*

```
stewiedev@DESKTOP-RH97L4T:~/lab2$ df .
Filesystem      1K-blocks    Used   Available Use% Mounted on
/dev/sdd        1055762868 1674268 1000385128  1% /
stewiedev@DESKTOP-RH97L4T:~/lab2$ df -k .
Filesystem      1K-blocks    Used   Available Use% Mounted on
/dev/sdd        1055762868 1674268 1000385128  1% /
stewiedev@DESKTOP-RH97L4T:~/lab2$ du -s *
4      error.txt
4      file1
4      file2
4      file3
4      merged.txt
4      yearfile
```